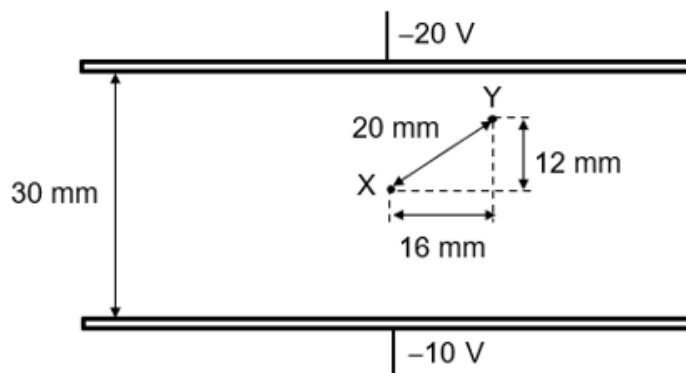


20. The diagram shows two large horizontal metal plates situated 30 mm apart. The top and bottom plates are at potentials -20 V and -10 V respectively. The magnitude of the electric field between the plates is 333 V m^{-1} .



What is the work done by the electric field when a charge of $-5.0 \mu\text{C}$ is moved from point X midway between the metal plates to point Y?

- | | | | |
|----------|---------------------------------|----------|---------------------------------|
| A | $-3.3 \times 10^{-5} \text{ J}$ | B | $-2.0 \times 10^{-5} \text{ J}$ |
| C | $2.0 \times 10^{-5} \text{ J}$ | D | $3.3 \times 10^{-5} \text{ J}$ |

Ans: B

The electric field $E = \frac{-10 - (-20)}{30 \times 10^{-3}} = 333 \text{ V m}^{-1}$ and it points upward (towards more negative plate)..

The electric force acting on the charge is $(5.0 \times 10^{-6})(333) = 1.67 \times 10^{-3} \text{ N}$, Since the charge is negative, the electric force is downward (opposite to electric field).

Work done by electric field $= qE \times d = (1.67 \times 10^{-3})(-0.012) = -2.0 \times 10^{-5} \text{ J}$.

Note: the distance moved parallel to the electric force is 12 mm, not 20 mm.

Also; the charge moved upward against the direction of the downward electric force, so the work done by the electric field should be negative.

21 In the following diagrams, the thin lines show equipotential lines and the bold arrows